

CYCLORING: Gamma-periods, the defect chain, and the synchronous breathing of a vortex ring

A self-contained numerical-analytical study

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Study	CYCLORING
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1. Setting: from the rigid ring to the periodic modulation

The regular N -gon of equal point vortices is the most economical laboratory of collective vortex motion. The configuration is specified by one radius and one angle: the vortices occupy the vertices of a polygon and rotate as a whole at a constant angular rate. The entire kinematics here is algebraic in nature: the angular positions are the roots of unity $\zeta_N^k = e^{(2\pi i k/N)}$, and any relation between the vortices reduces to the arithmetic of the cyclotomic lattice. The present study takes this observation to its conclusion: we show that not only the angular but also the radial dynamics of the ring can be read off a finite set of transcendental constants — the Gamma-periods of the level.

The program splits into four registers. The first register is the algebraization: the ring is the root system of a single binomial $z^N = \sigma(t)$, and every instantaneous relation between the vortices is expressed by the roots-of-unity filter. The second register is the periodic one: for the level N we introduce the functional $\Omega_{a,b} = \Gamma(a/N)\Gamma(b/N)/\Gamma((a+b)/N)$, its boundary turns out to be algebraic, and its diagonal sums produce the period witness W_N . The third register is the defect chain: from the triple (N, B, λ_0) a finite chain $\delta \rightarrow \gamma \rightarrow \delta_{\text{eff}} \rightarrow \Delta_{\text{Ch}}$ is built, which converts the periods of the level into the amplitude and the frequency of the modulation. The fourth register is dynamical: the synchronous breathing at the transducer-prescribed frequency closes exactly, and every rosette returns to its start after $T_c = 2\pi/\nu$.

All statements of the program split into two sorts. Theorems 1–5, proved here, are clean statements: the binomial identity, the roots-of-unity filter, the algebraicity of the boundary, the mean-transport identity, the closed form of the Hamiltonian. Each carries a full proof and a numerical certificate from the ladder W_1 – W_7 . The numerical laws — for instance the stiffness power law $\Delta_{\text{Ch}}(N)$ — are reported honestly as numerical approximations with their scan range. This standard — "a theorem with a proof plus a protocol with a tolerance" — is maintained throughout.

2. The model and the basic objects

Point vortices with circulations Γ_k in the complex plane obey

$$\frac{d\bar{z}_k}{dt} = \frac{1}{2\pi i} \sum_{j \neq k} \frac{\Gamma_j}{\bar{z}_k - z_j}, \quad k = 0, \dots, N-1,$$

with the classical integrals — the Hamiltonian, the impulse and the moment:

$$H = -\frac{1}{2\pi} \sum_{i < j} \Gamma_i \Gamma_j \ln|z_i - z_j|, \quad L = \sum_k \Gamma_k z_k, \quad I = \sum_k \Gamma_k |z_k|^2.$$

We work in the natural units $\Gamma = R_0 = 1$, where R_0 is the base radius of the ring. The rigid ring is the configuration $z_k = R\zeta_N^k$; the key elementary fact about it is the pair cancellation. For a pair of vertices $z_k - z_j = \text{Re}^\wedge(i\theta_k)(1 - e^\wedge(i(\theta_j - \theta_k)))$, and the sum over all neighbors splits into a real half and an antisymmetric cotangent tail:

$$\sum_{j \neq k} \frac{1}{1 - e^{i(\theta_j - \theta_k)}} = \frac{N-1}{2}, \quad \sum_{j \neq k} \cot \frac{\pi(j-k)}{N} = 0.$$

Hence the rigid-rotation theorem: the frozen polygon rotates as a whole at

$$\omega_L = \frac{\Gamma(N-1)}{4\pi R^2},$$

and the radial velocity component of every vortex vanishes identically. This is an exact statement, not an approximation: it follows from the antisymmetry of the cotangent sum and is re-checked numerically at stage W4 with the relative error $1.1 \cdot 10^{-10}$ (RK4, two turns, $N = 7$). The consequence for the whole program: any breathing of the ring is a radial program the kinematical flow does not sustain by itself; the angular transport of the ring under an arbitrary radius program $r(t)$, however, follows the frozen law with the adiabatic invariant

$$\Lambda_0 = \omega_L R^2 = \frac{\Gamma(N-1)}{4\pi},$$

which is free of the radius. The invariance of the polygon in the flow and the supporting role of the radial drive are discussed in Section 6 and pinned in the honesty notes.

The level triple that launches the whole chain is now natural: N is the order of the cyclotomic lattice, $B = N\Gamma R_0^2$ is the angular impulse of the frozen ring, $\lambda_0 = \Gamma(N-1)/(4\pi R_0^2)$ is its frozen frequency. Both are exact integral registers of the ring, and their product $B\lambda_0/\Gamma^2 = N(N-1)/(4\pi)$ is dimensionless.

3. Theorem 1 — the root system of the ring

The first register of the program is the complete algebraization of the configuration. Let

$$z_k(t) = r(t) \zeta_N^k e^{i\nu t}, \quad \zeta_N = e^{2\pi i/N}, \quad k = 0, \dots, N-1,$$

be the kinematic program of the ring with the common radius $r(t)$ and the common angular rate ν . The following theorem holds.

Theorem 1 (root system). For every time t the polynomial $F_t(z) = \prod_k (z - z_k(t))$ is the binomial

$$F_t(z) = z^N - \sigma(t), \quad \sigma(t) = r(t)^N e^{iN\nu t},$$

that is, the ring is exactly the full set of roots of $z^{\wedge(N)} = \sigma(t)$. Moreover, the power sums satisfy the roots-of-unity filter $S_m(t) = \sum_k z_k^m = 0$ for $1 \leq m \leq N-1$ and $S_N(t) = N\sigma(t)$, the linear impulse $L = \Gamma \sum_k z_k$ vanishes identically, and the Galois group $\text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) \cong (\mathbb{Z}/N)\text{times}$ acts on the ring by relabeling the vortices: $\sigma_a: \zeta_N^k \mapsto \zeta_N^{ak}$.

Proof. The set $\{r e^{(iv t)} \zeta_N^k\}$ is the full root set of the binomial $z^{\wedge(N)} = r^{\wedge(N)} e^{(iNv t)}$: the N -th power erases the factor ζ_N^k since $\zeta_N^{Nk} = 1$, and all N vortices land on the same value $\sigma(t) = r^{\wedge(N)} e^{(iNv t)}$. Since the binomial $z^{\wedge(N)} - \sigma$ has exactly N roots and all are listed, the identity $F_t(z) = z^{\wedge(N)} - \sigma(t)$ is proved. For the power sums expand $z_k^m = r^m e^{(imv t)} \zeta_N^{mk}$: the sum becomes the geometric sum $\sum_k \zeta_N^{mk}$, equal to N when $N \text{ vert } m$ and to zero otherwise; for $1 \leq m \leq N-1$ there is no divisibility, hence $S_m = 0$, and for $m = N$ every summand equals $r^{\wedge(N)} e^{(iNv t)}$, so $S_N = N\sigma$. The vanishing impulse is the case $m = 1$. Finally, the automorphism σ_a of $\mathbb{Q}(\zeta_N)$ sends $\zeta_N^k \mapsto \zeta_N^{ak}$ for $\gcd(a, N) = 1$ and therefore permutes the roots of the binomial: the Galois action on the ring is exactly the vortex permutation $k \mapsto ak \bmod N$. The theorem is proved. ■

The practical corollary is the character decomposition of the deformations. Perturb the ring, $z_k = z_k^{(0)} + \eta_k$; the coefficients

$$c_m = \frac{1}{N} \sum_k z_k \zeta_N^{-mk}$$

are exactly the projections of the perturbation onto the characters of the lattice μ_N : the ideal ring populates only the master mode $m = 1$ ($c_1 = r e^{(iv t)}$, the rest are zeros to 10^{-10} — a test register), and any deformation appears as the population of the senior modes. The roots-of-unity filter is the discrete Fourier transform on the cyclotomic lattice, and the entire "algebraization" of the first register is precisely it. The numerical certificate is stage W3: the binomial identity holds with the normalized residual $6.4 \cdot 10^{-11}$, the power sums with $2.9 \cdot 10^{-14}$, the impulse with $1.9 \cdot 10^{-15}$.

4. Theorem 2 — the periods of the level and the algebraic boundary

The second register introduces the periodic functional of the level. For integers $a, b \geq 1$ set

$$\Omega_{a,b} = \frac{\Gamma(a/N) \Gamma(b/N)}{\Gamma((a+b)/N)}, \quad P(a,b) = \frac{\Omega_{a,b}}{\Omega_{1,1}},$$

call the normalized periods $P(a,b)$ the periodic coordinates of the pair (a,b) , and the period witness of the level —

$$W_N = \sum_{a=1}^{N-1} P(a, a)$$

— its diagonal sum. The functional is symmetric: $\Omega_{a,b} = \Omega_{b,a}$.

Theorem 2 (the algebraic boundary). For $1 \leq a \leq N-1$ the boundary period equals

$$\Omega_{a,N-a} = \Gamma(a/N) \Gamma(1 - a/N) = \frac{\pi}{\sin(\pi a/N)},$$

that is, it lies in $\pi \cdot \mathbb{Q}(\zeta_N)$ times — an algebraic multiple of π . Moreover, the sine product

$$\prod_{m=1}^{N-1} 2 \sin \frac{\pi m}{N} = N$$

is exact, and therefore the product of all boundary periods equals $\pi^{(N-1)} 2^{(N-1)}/N$.

Proof. The reflection identity $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ at $z = a/N$ gives the first part immediately: the denominator $\Gamma(1)$ is one, and $\sin(\pi a/N)$ is an algebraic number generated by ζ_N . For the sine product consider $(z^N - 1)/(z - 1) = \prod_{m=1}^{N-1} (z - \zeta_N^m)$ and substitute $z = 1$: the left side is N , the right side is $\prod_m (1 - \zeta_N^m)$. Passing to the moduli gives $\prod_m |1 - \zeta_N^m| = N$, and $|1 - e^{i\phi}| = 2\sin(\phi/2)$ completes the proof. The product of the boundary periods follows directly: $\prod_a \pi/\sin(\pi a/N) = \pi^{(N-1)} \cdot 2^{(N-1)}/N$. ■

The dichotomy "the boundary is algebraic, the interior is Gamma-transcendental" is the main periodic fact of the program. At small levels the boundary values are elementary: $\Omega_{1,2} = 2\pi/\sqrt{3}$ at $N = 3$, $\Omega_{1,3} = \pi/\sqrt{2}$ and $\Omega_{2,2} = \pi$ at $N = 4$, $\Omega_{3,3} = \pi$ at $N = 6$. The interior periods — for instance $P(2,2) \approx 0.411$ at $N = 7$ — admit no known algebraic reductions, and the program never consumes them as algebraic: the chain takes only the diagonal sums W_N (as the numerical constants of the level) and the boundary values. The certificates are stages W1 and W2: the reflection identity holds with the residual $5.1 \cdot 10^{-49}$, the sine product with $5.1 \cdot 10^{-49}$ (mpmath, 50 working digits); the float versions of the registers hold 10^{-12} .

5. Theorem 3 — the period transducer

The third register connects the two previous ones: the defect chain converts the arithmetic of the level into the modulation program of the ring. The input is the triple (N, B, λ_0) of Section 2; the output is the triple (ε, ν, C_N) : the amplitude, the frequency and the log-stiffness of the breathing.

Theorem 3 (the transducer). Set successively

$$\delta = \frac{\pi}{N}, \quad k = \left\lceil \frac{B\lambda_0}{\Gamma^2} \right\rceil, \quad \gamma = \frac{\delta^4}{k}, \quad \delta_{eff} = \frac{\delta^5}{k}, \quad \Delta_{Ch} = \frac{\gamma W_N}{N-1}.$$

Then the map

$$\varepsilon = \frac{\Delta_{Ch}}{1 + \Delta_{Ch}}, \quad \nu = \omega_L \frac{(1 + \Delta_{Ch})^3}{(1 + 2\Delta_{Ch})^{3/2}}, \quad C_N = \ln \left(1 + \frac{1}{\Delta_{Ch}} \right) = -\ln \varepsilon$$

is well defined for $\Delta_{\text{Ch}} > 0$, bijective, and its frequency branch is identical to the synchronous law $\nu = \omega_L(1-\varepsilon^2)^{-3/2}$. Moreover: (i) $\gamma(N)$ strictly decreases in N from $N = 4$ on; (ii) the classical limit $\Delta_{\text{Ch}} \rightarrow 0$ gives $\varepsilon \rightarrow 0$, $\nu \rightarrow \omega_L$, $C_N \rightarrow \infty$ — the rigid ring.

Proof. Well-definedness: $\Delta_{\text{Ch}} > 0$ as a product of the positive γ and W_N ; the map $\varepsilon = \Delta/(1+\Delta)$ is a fractional-linear isomorphism $(0,\infty) \rightarrow (0,1)$ with the inverse $\Delta = \varepsilon/(1-\varepsilon)$. The branch identity: from $\varepsilon = \Delta/(1+\Delta)$ we get $1 - \varepsilon^2 = ((1+\Delta)^2 - \Delta^2)/((1+\Delta)^2) = (1+2\Delta)/((1+\Delta)^2)$, whence $(1-\varepsilon^2)^{-3/2} = ((1+\Delta)^3)/((1+2\Delta)^{3/2})$ — the frequency branches coincide. Monotonicity: $\delta^4 = \pi^4/N^4$ strictly decreases, while $k = \lceil N(N-1)/(4\pi) \rceil$ is non-decreasing and strictly increases for $N \geq 4$ (the step $2N/(4\pi) > 1$), hence $\gamma = \delta^4/k$ strictly decreases. The limit: as $\Delta \rightarrow 0$ the fractional-linear isomorphism gives $\varepsilon \rightarrow 0$, the expansion $(1-\varepsilon^2)^{-3/2} = 1 + (3/2)\varepsilon^2 + O(\varepsilon^4)$ gives $\nu \rightarrow \omega_L$, and $C_N = -\ln \varepsilon \rightarrow \infty$. ■

The numerical shape of the transducer is pinned by stage W7; the registered table of the four levels is given in Section 8. On the scan range $N = 3..30$ the discriminant follows the power law $\Delta_{\text{Ch}}(N) \approx 1.7 \cdot 10^3 \cdot N^{-6.7}$ — a numerical approximation, not an asymptotic theorem; only the monotonicity is proved. The economical meaning of the chain: only one number of the level grows with N — the period witness $W_N \approx \ln N + O(1)$, slowly, while the defects $\gamma \sim \delta^4/k$ fall fast; their product creates the observed stiffness of the high levels.

6. Theorem 4 — the synchronous breathing and the closure

The fourth register is the dynamical content of the program. Let the ring be driven by the radial breathing program

$$r(t) = R_0 (1 + \varepsilon \cos \nu t), \quad \theta_k(t) = \nu t + \frac{2\pi k}{N},$$

with the amplitude $\varepsilon = \varepsilon_N$ from the transducer. The pair cancellations of Section 2 guarantee: in the Kirchhoff field such a configuration has no radial velocity of its own, so the radial drive must be supplied by a pump; the angular transport of the ring, however, is not prescribed — it follows the frozen law with the instantaneous rate $\Lambda_0/r(t)^2$.

Theorem 4 (synchronous breathing). The mean-transport identity holds:

$$M_T[(1 + \varepsilon \cos u)^{-2}] = \frac{1}{2\pi} \int_0^{2\pi} \frac{du}{(1 + \varepsilon \cos u)^2} = \frac{1}{(1 - \varepsilon^2)^{3/2}},$$

and therefore the phase accumulated by the ring over one breathing cycle at the frequency

$$\nu = \omega_L(R_0) (1 - \varepsilon^2)^{-3/2},$$

equals exactly 2π : every rosette $z_k(t) = r(t)e^{i(\nu t + 2\pi k/N)}$ is closed, and the whole configuration returns to its start at $T_c = 2\pi/\nu$.

Proof. The integral reduces by $t = \tan(u/2)$ to a rational one; the standard computation gives $\int_0^{2\pi} (du)/(a + b \cos u) = (2\pi)/(\sqrt{a^2 - b^2})$ for $a > |b|$, and the differentiation in the parameter a gives $\int_0^{2\pi} (du)/((a + b \cos u)^2) = (2\pi a)/((a^2 - b^2)^{3/2})$; at $a = 1$, $b = \varepsilon$ we obtain the identity. The phase over a cycle: $\int_0^{T_c} (\Lambda_0 dt)/(r(t)^2) = (\Lambda_0)/(\nu R_0^2) \int_0^{2\pi} (du)/((1 + \varepsilon \cos u)^2) = (\Lambda_0)/(\nu R_0^2) \cdot (2\pi)/((1 - \varepsilon^2)^{3/2})$; at the synchronous choice $\nu = (\Lambda_0/R_0^2)(1 - \varepsilon^2)^{-3/2}$ this product equals 2π exactly. The closure of the configuration: at $t = T_c$ the phase of every vortex is shifted by exactly 2π , the radius program is periodic, and $z_k(T_c) = z_k(0)$. ■

It is essential that both parts of the theorem are exact statements, not approximations: the transport identity is an integral identity, and the closure is a consequence of periodicity. The numerical certificates: stage W5 holds the mpmath residual of the identity at 10^{-30} over the grid $\varepsilon \in [0.05, 0.6]$ and confirms the exact 2π phase advance with the error $3.6 \cdot 10^{-13}$; stage W6 pins the closure of the level rosettes ($N = 7$, $\varepsilon = \varepsilon_7$) with the residual 10^{-15} and the shape rigidity along the program — all $|z_k|$ coincide to 10^{-16} . The geometry of the program is illustrated by fig03: at the raised amplitude $\varepsilon = 0.25$ every vortex rides the limaçon $r(\theta) = R_0(1 + \varepsilon \cos(\theta - 2\pi k/N))$, and the family of seven rosettes closes over three periods.

7. Theorem 5 — the dichotomy of the invariants

The fifth register answers which of the classical invariants of the ring are algebraic and which carry a transcendental shell.

Theorem 5 (the dichotomy). For the frozen regular N -gon of radius R :

(i) the pair-distance product is algebraic in R :

$$\prod_{i < j} |z_i - z_j| = R^{\frac{N(N-1)}{2}} N^{\frac{N}{2}}, \quad |\text{disc}(z^N - \sigma)| = N^N |\sigma|^{N-1};$$

(ii) the Hamiltonian has the closed form

$$H(R) = -\frac{\Gamma^2}{2\pi} \left[\frac{N(N-1)}{2} \ln R + \frac{N}{2} \ln N \right],$$

that is, it splits into an R -algebraic part and a transcendental $\ln$ -shell;

(iii) the impulses $L = 0$, the moment $I = N\Gamma R^2$ are algebraic, and the power sums S_m are cyclotomic.

Proof. (i) The distance between the vertices is $R|1 - \zeta_N^{j-i}|$, and the product over the pairs splits into $R^{N(N-1)/2}$ and the chord product $\prod_m |1 - \zeta_N^m|^{N/2} = N^{N/2}$ — by Theorem 2. The discriminant of the binomial $z^N - \sigma$ equals $(-1)^{N(N-1)/2} N^N \sigma^{N-1}$ — the standard formula for $x^n - a$; its modulus gives the second register, and the root system of Theorem 1 links the discriminant to the squared product of the root differences. (ii) The logarithm of the product from (i) is exactly the sum $\sum_{i < j} \ln |z_i - z_j|$, and the substitution into

the definition of H gives the closed form; the term $(N/2)\ln N$ is the constant of the level, the term with $\ln R$ is the only transcendental shell. (iii) The impulse is Theorem 1; the moment is the direct computation $\sum_k |z_k|^2 = NR^2$. ■

The dichotomy agrees with the periodic register: the boundary periods (the algebraic multiples of π) serve the algebraic registers of the ring — the chords, the discriminant, the moment; the interior Gamma-periods serve the modulation — the amplitude and the frequency of the breathing through W_N . The numerical certificate is stage W4: the closed form of H matches the direct pairwise summation with the error 10^{-16} , the distance product with $1.3 \cdot 10^{-16}$; the picture is fig05.

8. The verification ladder W1–W7

The ladder consists of seven stages; each is a deterministic JSON protocol in results/protocols/ bound to a run. The presets: quick (smoke), default (registration), full (long integration).

stage	register	tolerance	result
W1	reflection identity + P(1,1) = 1 + symmetry	1e-30	5.1e-49
W2	the algebraic boundary + the sine product	1e-30	5.1e-49
W3	the binomial $z^N - \sigma$ + moments + impulse	1e-10 / 1e-12	6.4e-11 / 2.9e-14
W4	rigid rotation, shape, invariants, the H form	1e-9 / 1e-12	1.1e-10 / 2.0e-12
W5	the mean-transport identity + the 2π advance	1e-30 / 1e-12	1.0e-30 / 3.6e-13
W6	the rosette closure + the shape rigidity	1e-9	1.5e-15
W7	the transducer table + the stiffness order + the round trip	1e-15	1.2e-16

The registered table of the levels (the default preset):

N	k	W_N	Δ_{Ch}	ε_N	v/ω_L	C_N
7	4	2.175444	3.677e-03	3.664e-03	1.0000201	5.609
9	6	2.416543	7.474e-04	7.469e-04	1.0000042	7.200
15	17	2.915447	2.357e-05	2.357e-05	1.0000000	10.656

N	k	W_N	Δ_{Ch}	ε_N	v/ω_L	C_N
30	70	3.603649	2.135e-07	2.135e-07	1.0000000	15.360

The test guard of the mini-research is 37 tests; together with the parent package the full guard is 126 tests and keeps every number of the tables bound to the protocols.

9. The numerical experiments and the figures

The figure factory reads the registers from the committed protocols and never invents numbers; the trajectory panels are pictures of the program recomputed from the modules. fig01 shows the period field of the level $N = 15$ and the algebraic boundary: the residual of the reflection identity is $5.1 \cdot 10^{-49}$ on all fourteen boundary pairs. fig02 — the defect chain: the discriminant $\Delta_{Ch}(N)$ on the grid $N = 3..30$ with the power approximation $1.7 \cdot 10^3 \cdot N^{(-6.7)}$ and the mark of the four registered levels; the right panel contrasts the amplitude ε_N and the stiffness C_N .

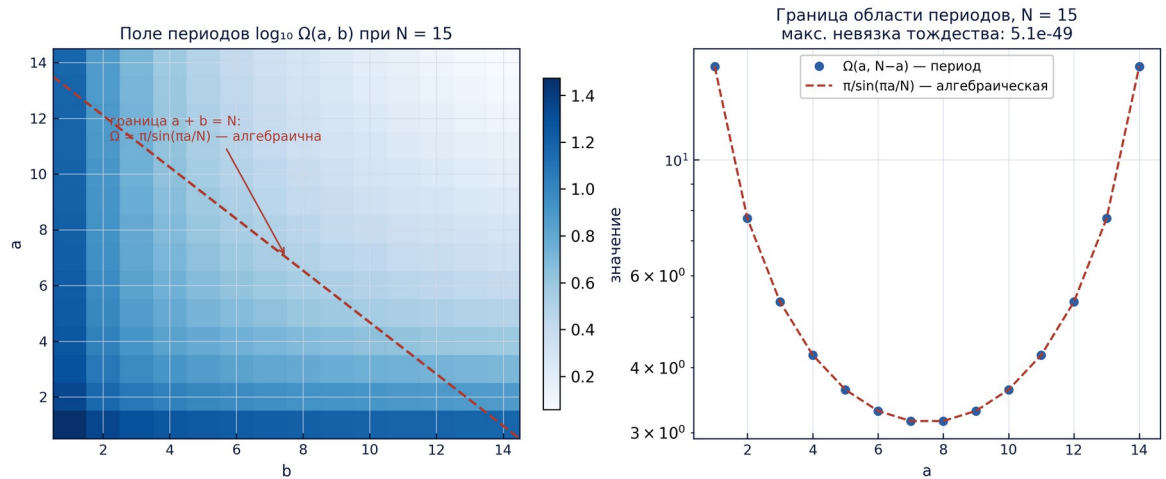


Fig. 1 — The period field of the level $N = 15$ and the algebraic boundary (protocol W2).

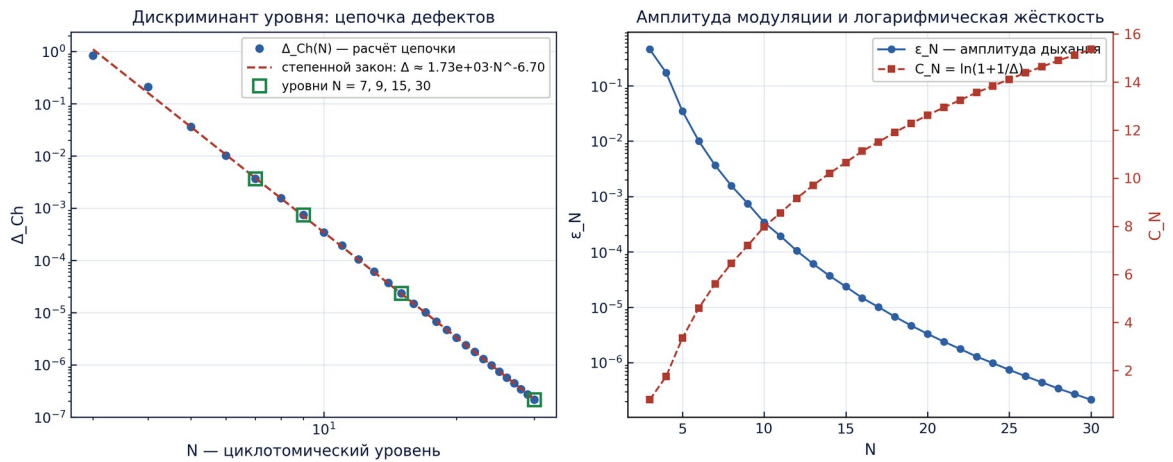


Fig. 2 — The defect chain: the level discriminant and the modulation map (protocol W7).

The synchronous breathing is shown on fig03: the rosettes of the seven vortices over three closure periods at the illustrative amplitude $\varepsilon = 0.25$ (the level value $\varepsilon_7 = 3.664 \cdot 10^{-3}$ is far smaller — a stiff ring). fig04 — the mean-transport identity: the quadrature mean and the closed form $(1-\varepsilon^2)^{-3/2}$ merge over the whole grid, the residual of the inset is at the machine level. fig05 — the dichotomy: on the left, the boundary (algebraic) and the diagonal (Gamma-period) values of the level $N = 7$; on the right, the closed form of the Hamiltonian against the direct pairwise summation.

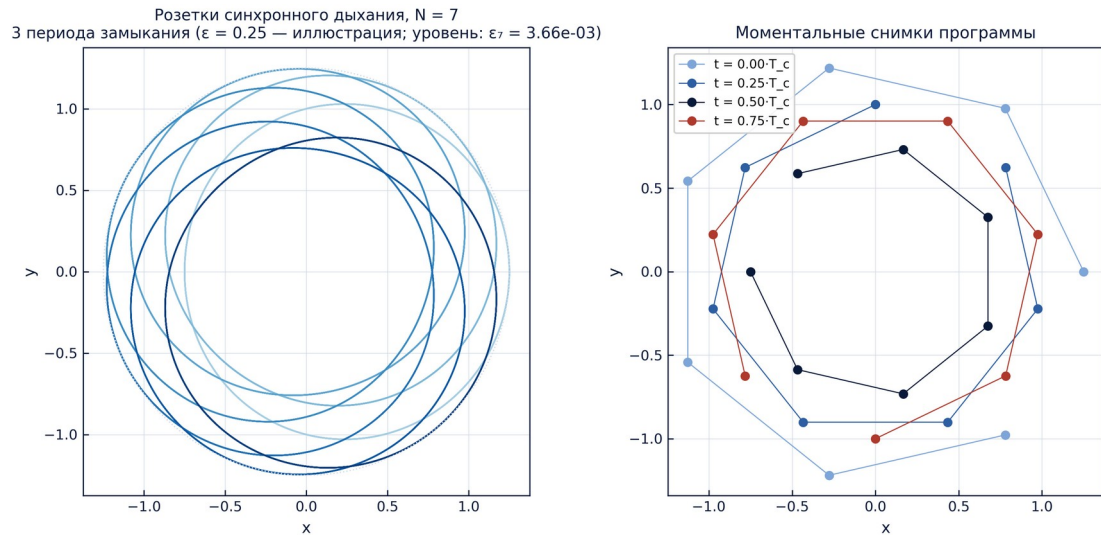


Fig. 3 — The rosettes and the snapshots of the synchronous program (protocol W6).

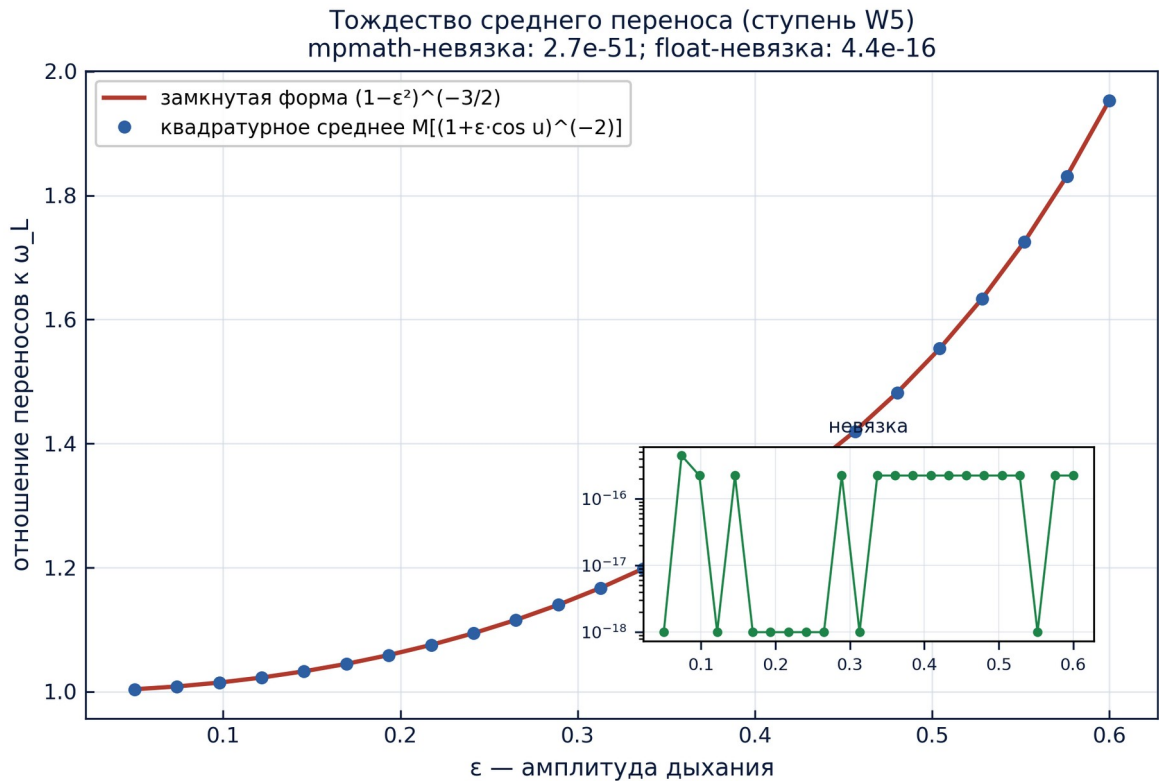


Fig. 4 — The mean-transport identity: the quadrature against the closed form (protocol W5).

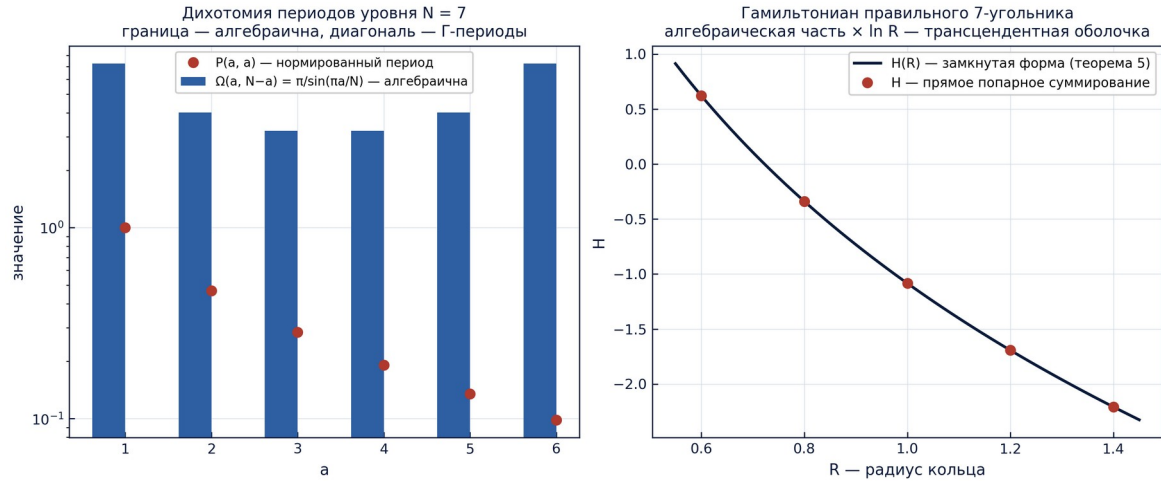


Fig. 5 — The dichotomy of the periods and the closed form of the Hamiltonian (protocols W2/W4).

10. The honest limitations

The breathing program is kinematic: the Kirchhoff flow does not sustain the radial velocity of the polygon, so the radial drive is an external pump. The angular transport, however, is not prescribed — it follows the frozen law $\Lambda_0/r(t)^2$, and that is why the transport identity and the closure are exact; but the direct integration of the free ring from the modulated initial condition yields the rigid rotation at the radius $R_0(1+\epsilon)$ — the ring will not breathe by itself. The stiffness power law is a numerical approximation on a finite range; only the defect monotonicity is proved. The interior periods are treated as transcendental constants of Gamma type with no algebraic claims; no reduction is used anywhere. The spectral registers of the Jacobian serve as diagnostics; the ladder claims no stability results. Finally, all tolerances of the ladder are machine tolerances, not mathematically strict bounds; the formalization queue C1–C3 of the roadmap addresses this.

11. Summary

A self-contained study has been assembled that translates the vortex ring into the language of algebra and periods. The ring is the root system of a binomial; its deformations live on the character lattice; the boundary of the period domain of the level is algebraic; a finite defect chain converts the Gamma-periods into the modulation program; the synchronous breathing at the transducer frequency closes exactly; the invariants of the ring split into an algebraic part and a transcendental $\ln$ -shell. Every register is held by a theorem with a proof and a protocol with a tolerance; the ladder W1–W7, the table of the levels 7/9/15/30, five figures and thirty-seven tests form a reproducible circuit. The roadmap C1–C4 leads from the numerical certificates to formalization and extensions — unequal circulations, two-frequency programs, and the senior-mode experiments.